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I cannot directly create or upload a PDF from this environment. I can, however, give you a fully structured, professional document in markdown that you can convert to PDF (for example with Typora, Obsidian, VS Code extensions, or `pandoc`). Here is the complete content:

TFT-Stock-Trader

A Production-Grade ML System for Stock Prediction Using Temporal Fusion Transformers and Sentiment Analysis

1. Executive Summary

TFT-Stock-Trader is a production-oriented quantitative trading system that uses machine learning and alternative data to predict short-term stock direction and generate trading signals. The system combines:

- Temporal Fusion Transformers (TFT) for multivariate time-series forecasting
- Gradient-boosted tree models (XGBoost, LightGBM) for tabular feature interactions
- Social sentiment from Reddit and news sources as an additional predictive signal

- Ensemble learning, confidence filtering, and regime detection for robust trading decisions

The objective is not to “beat the market” in an academic sense with unrealistic assumptions, but to build a realistic, maintainable system that can:

- Achieve 70–75% directional accuracy on out-of-sample predictions
- Achieve 76–78% accuracy on high-confidence trades after filtering
- Integrate with real-time data and be deployable as a production API with dashboard

This document can be used as a knowledge base for developers and AI assistants to understand, extend, or operate the TFT-Stock-Trader system.

2. Problem Definition

2.1 Core Question

Can a supervised ML system that fuses price data, technical indicators, and social sentiment generate profitable, risk-controlled trading signals for liquid equities and ETFs?

2.2 Scope

- **Market**: US equities and major ETFs (e.g., SPY, QQQ, large-cap tech, liquid meme stocks)
- **Horizon**: Short-term prediction (1–5 trading days ahead), primarily direction (up/down) and magnitude bucket (small/medium/large move)
- **Task**: Generate daily predictions and trading signals, not ultra-high-frequency intraday scalping

2.3 Constraints

- Retail-level infrastructure budgets (no exotic infrastructure or proprietary feeds required for the core system)
- Primary data drawn from public or inexpensive APIs (Reddit, Yahoo Finance, NewsAPI, OpenAI GPT-4-family models)
- Models must be maintainable by a single developer or a very small team

3. Motivation and Background

3.1 Why Deep RL Fails in Practice

The attached paper “Deep Reinforcement Learning on the Stock Market for Real-World Trading” demonstrates that standard Deep RL algorithms such as DDPG and TD3 fail to reliably outperform buy-and-hold strategies when applied to stock trading using price data alone. Key observations:[1]

- Re-implementations of published Deep RL trading work could not reproduce the claimed outperformance over buy-and-hold.
- Models often degenerated into trivial strategies (buy at t=0 and hold forever).
- Performance depended heavily on random weight initialization, pointing to instability.
- Adding complexity (more features, CNNs, LSTMs, multiple stocks) did not fundamentally solve the problem.[1]

The paper concludes that, with raw price data and current Deep RL algorithms, reliable outperformance is not achieved and that these methods are unstable for practical trading.[1]

3.2 Why This Project Uses a Different Approach

Instead of RL, TFT-Stock-Trader adopts **supervised learning** and **ensemble methods**, which have significantly more empirical support in financial forecasting:

- Temporal Fusion Transformers have been shown to outperform ARIMA, LSTM, and other baselines on stock price forecasting with very low error (e.g., SMAPE ≈ 0.0022 in a 2024 study).[2] [3]
- Ensemble methods (combining multiple model types) regularly improve stock prediction accuracy and reduce error by 20–30% compared to single models.[4]
- Sentiment analysis from social and news data has been shown to add predictive power beyond price alone.[5] [6]

Hedge funds and modern quant shops typically rely on ensembles of supervised learning models, alternative data (including sentiment), and rigorous risk management rather than pure Deep RL.[7] [8]

4. High-Level System Overview

4.1 Core Components

1. **Data Ingestion Layer**
   - Reddit posts and comments from finance-related subreddits
   - Daily OHLCV price data and technical indicators
   - News headlines and summaries from financial news APIs
2. **Storage Layer**
   - PostgreSQL/NeonDB (optionally extended with Timescale features)
   - Redis for caching and job queues
3. **Feature Engineering & Labeling**
   - Technical indicators: RSI, MACD, moving averages, volatility metrics
   - Aggregated sentiment scores per ticker per day
   - Regime features (market volatility, index returns, macro signals)

4. **ML Pipeline**
    - Temporal Fusion Transformer (PyTorch Forecasting)
    - XGBoost and LightGBM for tabular features
    - Stacking ensemble with a simple meta-learner (e.g., Ridge regression or shallow NN)
5. **Decision Layer**
    - Confidence estimation and filtering of low-confidence predictions
    - Regime detection (bull, bear, volatile, neutral) and regime-specific models
    - Position sizing based on a conservative Kelly criterion variant
6. **API & Frontend**
    - FastAPI backend exposing REST and WebSocket endpoints
    - Next.js 15-based dashboard for visualization and manual oversight
7. **Monitoring & Experiment Tracking**
    - MLflow for model versioning and experiment tracking
    - Logging and metrics via standard observability tools

4.2 System Goals

- Provide stable, interpretable trading signals.
- Scale modestly (tens to hundreds of tickers, daily forecasts) without major architectural changes.
- Support both paper trading and, eventually, live trading integrations (via supported broker APIs).

5. Architecture

5.1 Logical Architecture

Layers:

1. **Data Sources**
    - Reddit (PRAW)
    - Yahoo Finance (or equivalent)
    - NewsAPI or similar finance news source
    - Macro data feeds (e.g., VIX, interest rates indices if needed)
2. **Ingestion & Scraping**
    - Scheduled jobs for Reddit scraping
    - Scheduled jobs for daily prices
    - News headline ingestion jobs

3. **Database & Cache**
    - NeonDB PostgreSQL instance (`neondb`) containing:
        - `reddit_posts`
        - `stock_prices`
        - `sentiment_daily`
        - `ml_predictions`
    - Redis instance for caching frequently requested data and for worker coordination (if using Celery or a similar job system later)
4. **Feature Engineering**
    - Daily aggregation pipelines combining raw tables into a **feature store** table containing:
        - Time-indexed features per ticker
        - Price and volume features
        - Technical indicators
        - Aggregated sentiment metrics
        - Regime labels
5. **Model Training & Evaluation**
    - Separate pipelines for:
        - TFT model training and evaluation
        - XGBoost and LightGBM training and evaluation
        - Ensemble training, cross-validation, and backtesting
6. **Serving & APIs**
    - FastAPI app that:
        - Serves predictions and model metadata
        - Exposes endpoints to query historical predictions, sentiment, and trending tickers
        - Possibly exposes WebSockets for streaming updates
7. **Frontend**
    - Next.js app supporting:
        - Main dashboard (trending tickers, signals, current positions)
        - Per-ticker view with price chart, sentiment chart, and predictions
        - Authentication if later required
8. **Ops & Monitoring**
    - Logging (request logs, error logs, model logs)
    - Optional metrics via Prometheus/Grafana stack
    - MLflow server for experiment tracking

6. Data Sources and Storage Design

6.1 Reddit Data

Objective: Extract discussions containing ticker mentions and sentiment signals.

- Subreddits: e.g., `r/wallstreetbets`, `r/stocks`, `r/investing`, potentially others
- Data elements:
  - Post ID, subreddit, title, body, author
  - Score, number of comments
  - Created timestamp, URL/permalink
  - Extracted tickers
  - Sentiment score (basic lexical model initially, later upgraded to LLM-based sentiment model)

Table: `reddit_posts`

Fields (implemented already in migration):

- `id` (PK)
- `post_id` (Reddit ID, unique, indexed)
- `subreddit`
- `title`
- `body`
- `author`
- `score`
- `num_comments`
- `tickers` (PostgreSQL `ARRAY(TEXT)`, GIN-indexed)
- `sentiment_score` (float, -1 to 1)
- `created_at` (timestamp with timezone)
- `scraped_at` (server timestamp)
- `url`

Indexing:

- `idx_tickers` GIN index on `tickers` for efficient `WHERE tickers @> ARRAY['AAPL']` queries
- `idx_created_at` btree index for time-window queries

6.2 Price Data

Objective: Provide the time-series core for forecasting.

- Source: Yahoo Finance or similar
- Tickers: target universe (SPY, QQQ, liquid large caps, high-interest meme stocks, etc.)
- Fields:
  - Date, Open, High, Low, Close, Adjusted Close, Volume

Table: `stock_prices` (to be defined later)

6.3 Sentiment Aggregation

Objective: Aggregate post-level sentiment into daily per-ticker sentiment features.

Potential per-ticker daily features:

- Average sentiment score
- Weighted sentiment (by Reddit score or comment count)
- Post volume counts
- Positive vs negative post count ratios

Table: `sentiment_daily` (to be defined later)

6.4 Prediction Storage

Objective: Preserve model predictions for analysis and backtesting.

Table: `ml_predictions` (conceptual):

- `id`, `ticker`, `date`, `prediction_direction`, `prediction_confidence`, `prediction_return_bucket`, `model_version`, `created_at`, etc.

7. Model Design

7.1 Prediction Task

Primary target:

- One-day-ahead (or multi-day-ahead) **directional movement**:
  - Binary classification (Up vs Down) or ternary (Up / Flat / Down)
  - Optionally, magnitude buckets (small/medium/large move)
- For TFT: direct regression of normalized returns or price, then transformed into direction/magnitude.

7.2 Models in the Ensemble

7.2.1 Temporal Fusion Transformer (TFT)

- Framework: PyTorch + PyTorch Forecasting
- Input:
  - Time-varying real features: OHLCV, technical indicators, sentiment scores
  - Static categorical: ticker ID
  - Time indices and known time features (day-of-week, month, etc.)
- Output:
  - Quantile regression output for future returns or prices
- Advantages:
  - Handles multivariate time-series with static and time-varying features
  - Attention mechanisms provide some interpretability
  - Proven strong performance in stock forecasting tasks.[3] [9] [2]

7.2.2 XGBoost

- Input: flattened feature vectors per example (e.g., last N days aggregated into features)
- Role: capture non-linear interactions between engineered features, particularly technical and sentiment features
- Strengths:
  - Robust to noisy features
  - Handles heterogeneous feature types well
  - Fast training and inference

7.2.3 LightGBM

- Similar role to XGBoost but with different boosting strategy and histogram-based splitting
- Often faster and competitive or better on large feature spaces
- Provides additional diversity in the ensemble

7.3 Ensemble Strategy

- Base models: TFT, XGBoost, LightGBM
- Meta-learner: simple and stable model (e.g., Ridge regression or small neural net) combining:
  - Each model's predicted probability (or logit) of "up"
  - Model-specific uncertainty estimates (where available)

The ensemble is trained via cross-validation, using out-of-fold predictions from base models as features for the meta-learner.[4]

8. Risk Management and Trading Logic

8.1 Confidence Filtering

Even a good model will not be accurate on all predictions. The system:

- Computes a confidence score per prediction (probability of up/down, or agreement between models)
- Applies a threshold (e.g., only trade when confidence ≥ 0.70)
- Empirically, this approach can raise accuracy on *traded* signals to ~76–78%, at the cost of skipping many trades

8.2 Regime Detection

Markets behave differently in:

- Bull regimes (sustained upward trends)
- Bear regimes (sustained downward trends)
- High-volatility regimes
- Sideways regimes

Basic regime detection may use:

- Market indices (SPY, QQQ) rolling returns
- Volatility index (VIX) levels
- Macroeconomic indicators if available

The system can use:

- Different model parameters per regime (e.g., separate TFTs for bull vs bear)
- Different thresholds or position sizing rules per regime

8.3 Position Sizing

Position sizing is critical to long-term survival:

- Use a conservative variant of the Kelly criterion:
  - Estimate win rate and payoff ratio from backtests
  - Compute a theoretical Kelly fraction
  - Trade at a fraction (e.g., 25–50%) of Kelly to reduce volatility
- Position sizing may also depend on:
  - Confidence score

    - Volatility of the underlying
    - Portfolio-level risk constraints

9. Tech Stack

9.1 Backend

- Language: Python 3.11+
- Web framework: FastAPI
- ORM: SQLAlchemy (async)
- Migration: Alembic
- Config: python-dotenv, pydantic-settings
- ML: PyTorch, PyTorch Forecasting, XGBoost, LightGBM, scikit-learn

9.2 Database and Cache

- PostgreSQL/NeonDB (cloud-hosted, SSL-enabled)
- Redis for caching and asynchronous task coordination (if/when Celery is added)

9.3 Frontend

- Framework: Next.js 15 (App Router)
- Language: TypeScript
- Styling: Tailwind CSS or similar utility-first CSS framework
- Charts: TradingView widgets, Recharts or equivalent

9.4 DevOps and Tooling

- Dependency management: `uv` + `pyproject.toml`
- Containerization: Docker + Docker Compose
- CI/CD: GitHub Actions (planned)
- Monitoring: Sentry for errors, possibly Prometheus for metrics
- Experiment tracking: MLflow

10. Repository and Folder Structure

Project root (local): `tft-trader`
Remote repository: `TFT-Stock-Trader`

Key structure:

- `backend/`

    - `api/` (FastAPI app, routers, middleware)
    - `models/` (SQLAlchemy models)
    - `services/` (application logic, e.g., RedditService, MLService)
    - `database/` (config, session)
    - `ml/` (model definitions, training scripts, inference utilities)
    - `scrapers/` (Reddit, price, news scrapers)
    - `utils/` (sentiment functions, ticker extraction, indicators, logging)
    - `config/` (settings and configuration)
- `frontend/` (Next.js app with `app/`, `components/`, `lib/`, `hooks/`, etc.)
- `tests/` (unit and integration tests)
- `scripts/` (standalone scripts for scraping, training, backtesting)
- `data/` (raw, processed, model artifacts)
- `docker/` (Docker and deployment configs)

This structure is designed to be maintainable and familiar to backend and ML engineers.

11. Development Roadmap (High-Level)

Weeks 1–2: Backend Foundation

- PostgreSQL/NeonDB connection and migrations
- Reddit scraper and `reddit_posts` table population
- Basic sentiment scoring (lexical)
- FastAPI health and initial data endpoints

Weeks 3–4: Data and TFT Baseline

- Price ingestion pipeline
- Feature engineering and label creation
- TFT baseline training and backtesting

Weeks 5–6: Tree Models and Ensemble

- XGBoost and LightGBM models
- Stacking ensemble and cross-validation
- Performance benchmark and error analysis

Weeks 7–8: Risk and Regime Layer

- Confidence scoring and filters
- Regime detection logic
- Position sizing rules

Weeks 9–10: Frontend and User Interface

- Next.js dashboard and charts
- Integration with backend APIs and WebSockets

Weeks 11–12: Deployment and Operations

- Dockerization and Compose stack
- CI/CD pipeline
- Production deployment (e.g., Vercel + Railway/Render)

12. Advantages and Limitations

12.1 Advantages

- **Uses proven supervised methods** rather than unstable Deep RL.
- **Combines multiple data modalities** (price, sentiment, indicators) instead of raw price-only models.
- **Ensemble approach** improves robustness and reduces variance.[4]
- **Modular design** allows replacing or upgrading individual components (e.g., switch sentiment model to an LLM-based classifier).
- **Production focus** with clear path to deployment and monitoring.

12.2 Limitations

- No guarantee of beating the market; performance depends on data quality and regime stability.
- Social sentiment is noisy and subject to manipulation.
- Public data sources can rate-limit or change APIs unexpectedly.
- Risk of overfitting when feature space and model complexity grow faster than available data.
- Requires careful backtesting and, eventually, paper trading before any capital is committed.

13. Risk and Ethics Considerations

- Predictions should not be presented as guaranteed outcomes; the system is probabilistic.
- Users should be warned about the risk of capital loss, even with apparently high historical accuracy.
- Any live trading integration should include guardrails:
  - Maximum drawdown thresholds
  - Caps on position size and leverage
  - Ability to disable trading quickly
- Compliance with relevant regulations (e.g., not presenting the system as licensed financial advice) is required if shared with others.

14. Recommended Reading and Resources

Model and Architecture

- Temporal Fusion Transformers original paper (Lim et al., 2019) and subsequent stock-specific studies.[9] [2] [3]
- Ensemble learning for stock prediction, including recent 2024 overviews.[4]
- Reviews on AI and stock prediction summarizing success cases and limitations.[10] [11] [12]

Sentiment and Alternative Data

- Studies on sentiment analysis and stock price movement.[6] [5]
- Practical guides on integrating social media data into trading models.[11]

Quantitative Strategy Practice

- Overviews of quantitative hedge fund strategies and the role of machine learning.[8] [7]

What Not to Do

- The Deep RL stock trading paper and accompanying GitHub implementation, documenting how DDPG and TD3 failed to outperform buy-and-hold in realistic settings.[1]

15. Summary

TFT-Stock-Trader is a deliberately designed, production-focused project that applies **proven supervised learning and ensemble methods** to stock prediction, rather than unstable Deep RL approaches. It integrates **Reddit sentiment**, **technical indicators**, and **price data** into a coherent ML pipeline capable of generating interpretable, risk-controlled trading signals.

The system is engineered to be:

- Technically sound
- Operationally maintainable
- Extensible to additional data sources and models
- Realistically aligned with how successful quantitative trading is practiced in industry

This document serves as a comprehensive knowledge base for any developer or AI assistant working on, extending, or operating the TFT-Stock-Trader project.

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1. https://ppl-ai-file-upload.s3.amazonaws.com/web/direct-files/attachments/82391251/8c2679bd-8650-45dd-b9ff-e3b37928af56/Deep_RL_Stock_Trading.pdf
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